

SARAH KAYEMBE

sarahmbumbakayembe@gmail.com | (207) 347-0465 | linkedin.com/in/sarah-kayembe

EDUCATION

University of Southern Maine, Portland, ME - M.S. Computer Science (GPA 3.8) Graduated May 2026
University of Botswana, Gaborone, Botswana - B.S. Computer Science Graduated May 2023
Relevant Coursework: Artificial Intelligence, Machine Learning, Data Mining, Data Structures & Algorithms, Object-Oriented Programming, Software Testing, Web Technologies, API Development

TECHNICAL SKILLS

Python, JavaScript, Java, SQL, scikit-learn, XGBoost, Deep Neural Networks, Transformers, LLM Reranking, Text Classification, Prompt Engineering, Pandas, NumPy, Machine Learning, NLP, Django, React, Flask, FastAPI, REST APIs, HTML5, CSS3, Node.js, PostgreSQL, MySQL, Git, AWS, Linux, Automated Testing, Computer Vision, OpenCV, MongoDB, Next.js

EXPERIENCE

University of Southern Maine | Software Developer, STEM Outreach Jan 2024 – Present

- Developed and maintained the STEM Outreach web platform using WordPress, HTML5, and CSS3, implementing responsive UI improvements across 20+ pages to enhance accessibility, user engagement and overall user experience.
- Optimized website management workflows and content delivery processes, reducing update cycles by 40% while improving maintainability for university outreach operations.

University of Southern Maine | Computer Science Teaching Assistant Sep 2024 – Mar 2026

- Designed curriculum and taught undergraduate Java programming courses across 3 semesters, leading lectures and labs for 60+ students covering object-oriented programming, algorithms, analytical reasoning, debugging, and software engineering practices.
- Reviewed and evaluated 100+ student code submissions using testing and debugging methodologies, improving programming quality, documentation, and development practices.

Care For Humanity | Full-Stack Developer May 2023 – Dec 2025

- Engineered and launched a responsive production website using HTML5, CSS3, and JavaScript, building 10+ pages to improve digital accessibility, user experience, and organizational outreach.
- Implemented reusable frontend components and optimized website performance, creating a maintainable platform for future content updates and community engagement.

Wayside Publishing | Usability Testing Analyst Intern Jan 2024 – Dec 2024

- Executed usability testing for Nualang, identifying interface issues and reducing developer rework by 50% through early defect detection and structured feedback.
- Evaluated user behavior data and translated findings into UI recommendations, contributing to redesigned layouts achieving 98% user preference.

RESEARCH & PROJECTS

StudyBuddy Hub – Full-Stack Web Application (🔗 Github)

- Architected and implemented more than 20 RESTful API endpoints with ACID transactions, pagination, SQL injection prevention, and optimized database queries to deliver secure, scalable backend services.
- Built a production-style full-stack study platform using React, Flask, MySQL, and REST APIs to support quiz creation, flashcards, user authentication, and collaborative study features.
- Integrated reusable React components with Hooks, React Router, Axios, and Vite, connecting frontend interfaces with backend services through well-structured REST APIs.

Segmentation Granularity & LLM Reranking - Master's Thesis (🔗 Paper)

- Benchmarked a four-stage NLP pipeline in Python for legal argument retrieval, comparing segmentation strategies (document, paragraph, sentence, proposition), transformer models, and aggregation methods across 60+ configurations to isolate the effect of each design choice.
- Constructed a multi-stage retrieval and reranking architecture using BM25 for candidate retrieval, fine-tuned cross-encoders for classification, and zero-shot DeepSeek reranking to improve legal evidence matching, showing that a strong instruction-tuned LLM outperforms a fine-tuned cross-encoder baseline without task-specific training.

Employee Turnover Prediction - IEEE/ACIS SERA 2025 Publication (🔗 Paper)

- Applied machine learning to 778 employee survey responses by engineering features from job satisfaction, income, management perception, and tenure data.
- Trained Decision Trees, Random Forests, and Deep Neural Networks using cross-validation and feature importance analysis to predict turnover risk.
- Co-authored peer-reviewed research published at IEEE/ACIS SERA 2025 on ML-driven employee retention prediction.